Scalability & Future Plan

Date	7 July 2026
Team ID	XXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:
This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports small to medium user traffic.	Deploy using multiple Gunicorn workers behind Nginx or a cloud load balancer to handle concurrent users.
2	Data Storage	Uses CSV dataset and .pkl model files.	Integrate a relational database such as MySQL or PostgreSQL to store applicant information and prediction history.

3	Performance	Prediction is performed using a locally loaded XGBoost model.	Optimize inference, implement caching, and containerize the application using Docker for better

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	The application processes one prediction request at a time and is not optimized for a large number of concurrent users.	Performance may decrease under heavy traffic.	High
2	The system uses a CSV dataset and serialized model files instead of a database.	No support for storing user history or managing prediction records.	Medium
3	User authentication and role-based access control are not implemented.	Limits secure access and administrative functionality.	Medium

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Scalability Plan

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4	Security	Basic input validation is implemented.	Add user authentication, role-based authorization, HTTPS, secure session management, and encrypted sensitive data.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add user login, registration, applicant history, and database integration.	2–3 Months	Improve security, data management, and user experience.
Phase 3	Develop an admin dashboard with analytics, prediction reports, and applicant management.	3–4 Months	Provide better monitoring and decision support for financial institutions.

Phase 4	Deploy using Docker and cloud infrastructure, expose REST APIs, and integrate additional financial indicators for improved prediction accuracy.	4–6 Months	Enhance scalability, availability, and overall system capabilities.